Answers and talk notes

Represent Three-Dimensional Figures in Two Dimensions

What your student is learning

Tonight we look at a flat net and name the solid it folds into — by counting its pieces and their shapes.

Answers

Ask your student to explain before you look.

1. **B.** Two. Two: a prism has two matching bases.
2. **D.** One. One: a pyramid has one base and its other faces meet at a point.
3. **A.** Square pyramid. Square pyramid: one square base and four triangles that meet at a point.
4. **C.** Triangular prism. Triangular prism: two triangle bases and three rectangle sides.
5. **B.** Rectangular prism. Rectangular prism: all six faces are rectangles in matching pairs.
6. **D.** 5. 5 faces: 2 triangle bases and 3 rectangles.

Talk about it

How many bases does the net have, and what shape are they? What does that tell you about the solid?

Home mission

Open an empty box flat and point to its two matching bases. Can your student say how they know this net folds into a prism and not a pyramid?

When they are stuck, ask

- How many pieces match each other exactly?
- Does this solid have one base or two?
- What shape is the base? That goes in the name.

Try not to say

- Count one face from each pair, the rest are the same.
- Any net with triangles is a pyramid.
- The number of faces does not matter for naming it.